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Accelerated degradation of photovoltaic modules under a future warmer climate

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Abstract

Solar photovoltaic (PV) module deployment has surged globally as a part of the transition towards a decarbonized electricity sector. However, future climate change presents issues for module degradation due to prolonged exposure to outdoor conditions. Here, we identify key degradation mechanisms of monocrystalline-silicon (mono-Si) modules and empirically model their degradation modes under various climate scenarios. Modules tend to degrade faster due to the thermal degradation mechanism. We estimate that the weighted average degradation rate will increase up to 0.1%/year by 2059. On assessing the impacts of module degradation on future PV power generation and levelized cost of energy, we project up to 8.5% increase in power loss that leads to ~10% rise in future energy price. These results highlight the need to climate-proof PV module design through careful material selection and improvements in the module manufacturing process. In particular, we recommend the use of heat dissipation techniques in modules to prevent degradation due to overheating.

KEYWORDS

climate change, future projections, impacts of module degradation, module degradation

1 | INTRODUCTION

Deploying large-scale renewable energy technology is key to reducing global carbon emissions and mitigating climate change. Renewable energy integration plays a major role in achieving the United Nations Sustainable Development Goal 7 and contributing to energy security, climate change adaptation and resilience. Solar Photovoltaic (PV) technology is expected to become one of the largest sources of renewable energy worldwide by 2026.¹ Australia is quickly transitioning to include PV technology in its energy mix to meet the Renewable Energy Target. The installed PV capacity in Australia is expected to reach ~80 GW in 2030 from 30.5 GW in 2023.²

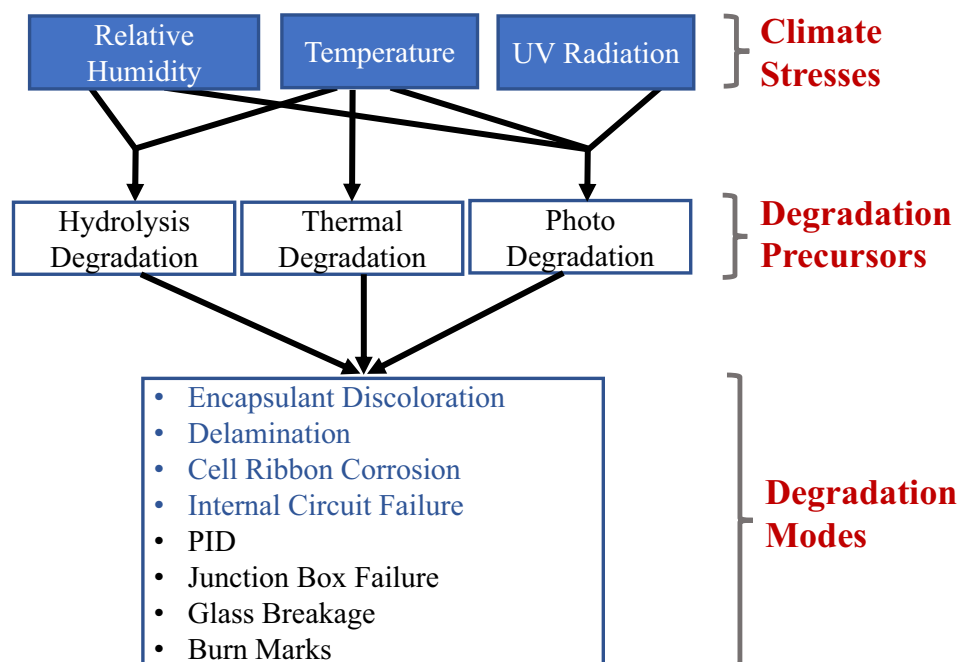
Exposure of PV modules under different environmental conditions influences the chances of module degradation and failure,³ which imposes module reliability issues. To safeguard the PV system performance, it is essential to have reliable and durable PV modules.⁴ Degradation of PV modules during their lifespan results in power loss⁵ and introduces financial risks. Inaccurate estimation of degradation rates can lead to inaccurate forecasts of future energy generation and hence the levelized cost of energy (LCOE). Therefore, it is essential to understand the mechanisms behind module degradation.

Most module manufacturers provide a 25- to 30-year linear warranty period, during which they guarantee power drops of less than 20%.^{4–6} However, extensive research has highlighted that the majority of modules start degrading during their warranty period due to

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FIGURE 1 Overview of the drivers of the degradation modes and mechanisms. The degradation modes in blue are discussed in this study since they have the highest occurrence probability.¹⁷



climatic conditions, material properties of different PV technology and poor quality during manufacturing.^{5,7-11} The degradation rates vary from 0.26%/year to 2.3%/year depending on the dominant degradation mechanism and module type.^{7,10} Degradation rates of silicon modules have been extensively studied for various regions of the world^{5,12-14} and have been reported to be on average ~0.5%/year.^{15,16} Previous studies^{15,16} have identified ultraviolet (UV) radiation, temperature and humidity as the main climate stresses that lead to different degradation mechanisms by themselves or in synergy. Kaaya et al.¹⁵ classify three important degradation precursors: thermo-mechanical degradation (caused due to high module temperature), photodegradation (caused due to temperature, UV radiation and humidity) and hydrolysis degradation (caused due to humidity and temperature). These degradation precursors can induce several degradation modes like delamination, encapsulant discoloration, potential-induced degradation (PID), internal circuit failure, cell crack, glass breakage and hot spots in PV modules. The climate stresses responsible for degradation precursors are shown in Figure 1. It also shows the degradation precursors responsible for inducing different degradation modes in PV modules. In this study, we only consider the degradation modes in blue (Figure 1) since they have the highest occurrence probability.¹⁷

Hot spots (33%) are the most common mode of degradation in monocrystalline-silicon (mono-Si) modules. The other common modes of degradation in mono-Si modules are ribbon discoloration (20%), glass breakage (12%), encapsulant discoloration (10%), cell breakage (9%) and PID (8%).^{6,17} In the past, Ascencio-Vásquez et al.¹⁶ determined the total degradation rate of crystalline-silicon modules by considering linear interactions of thermal degradation, hydrolysis degradation and photodegradation. However, their method does not account for the probability of occurrence of individual mechanisms. Considering the influence of climate on driving degradation

mechanisms, their probability of occurrence will also vary for different climate types. Hence, the total degradation rate based on the occurrence probability of different mechanisms and modes will vary regionally. At present, module degradation modes are not considered individually due to the difficulty in measuring the power from a single module in a PV system and estimating power loss due to individual degradation modes.^{5,7-11} Additionally, a cohesive statistical model based on various degradation modes to predict the power curve over time remains a challenge in the field of PV degradation modelling; there are however recent advancements in this field.¹⁸ Local weather and climate influence the degradation of PV modules. Hence, similar modules may experience different degradation rates depending on the installation site. Further, the dependency of module degradation mechanisms on meteorological parameters makes it highly susceptible to future climate change. Even though degradation rates have been studied for specific sites,^{7,11,15,16,19-21} to date, there has yet to be a study that predicts the future degradation modes and estimates future degradation rates spatio-temporally for Australia under different climate change scenarios. Considering the large-scale PV deployment in Australia, it is essential to estimate the future degradation rates accurately before investing.

In this research, we aim to understand the degradation mechanisms of mono-Si modules Australia-wide using high resolution climate model projections. Mono-Si PV technology is the dominant PV technology worldwide due to its cost-effectiveness.²² We estimate the module degradation modes for the historical (1976–2005) and future (2030–2059) periods under a low-emission scenario (representative concentration pathway [RCP] 2.6) and a high-emission scenario (RCP8.5) using regional climate model (RCM) projections from the Coordinated Regional Downscaling Experiment (CORDEX).^{23,24} For the first time, this research aims to statistically model the weighted average degradation rate Australia-wide for the different degradation

TABLE 1 The CORDEX-Australasia ensemble members analysed in this study.

	CCLM	REMO2015	RegCM4.7
MPI-ESM-LR			
MPI-ESM-MR			
NORES1-M			
HadGEM2-CC			

Note: Climate projections are used for the historical period (1976–2005) and future period (2030–2059) for RCP2.6 and RCP8.5 scenarios. The highlighted boxes indicate the available GCM-RCM pairs.

modes (delamination, internal circuit failure, cell ribbon corrosion and encapsulant discoloration). We also estimate the economic impacts of enhanced future degradation of mono-Si modules across Australia. Understanding dominant degradation mechanisms and modes for different locations across Australia will be highly beneficial for identifying the module designs most vulnerable to future degradation. Thus, they can be useful in improving the module design and thereby improving the durability of the modules.

2 | METHODS

2.1 | RCM projections

In this study, regional climate model (RCM) projections from the CORDEX for Australasia²⁵ are used to study the degradation of PV modules over Australia. Climate model projections over the CORDEX-Australasia region are obtained by downscaling four global climate models (GCMs) from the Coupled Model Intercomparison Project—Phase 5 (CMIP5) using three RCMs. This study uses all ensemble members that simulate the required emission scenarios, giving us a nine-member ensemble to understand module degradation in historical and future periods (Table 1). We estimate the future degradation rates for a low-emission scenario (RCP2.6) and a high-emission scenario (RCP8.5). The 3-hourly CORDEX-Australasia simulations are used to study the historical (1976–2005) and future (2030–2059) periods. We have used temperature and relative humidity from CORDEX-Australasia simulations to obtain thermal and hydrolysis-driven degradation mechanisms. We have obtained UV radiation using the method proposed by Wald.²⁶ This method uses global horizontal irradiance (GHI) from CORDEX-Australasia and clearness index (K_t) estimated using the open-source Python library PVLIB.²⁷

Previous studies have evaluated the robustness of CORDEX-Australasia ensembles by comparing the historical projections with observations and reanalysis datasets. These studies have reported minimum bias in variables like radiation, minimum temperature, maximum temperature and precipitation.^{25,28–30} This adds confidence in using CORDEX-Australasia projections to estimate PV module degradation across Australia.

2.2 | Degradation mechanisms

This section presents details of the various degradation mechanisms observed in mono-Si modules (with glass back sheet) due to climate stresses. Photodegradation, thermal degradation and hydrolysis degradation are identified as the three main degradation precursors.^{15,16} These precursors lead to various degradation modes.

2.2.1 | Hydrolysis degradation

Relative humidity and temperature are the primary climate stresses that initiate hydrolysis degradation mechanism. Empirically hydrolysis-driven degradation can be modelled using the Pecks model,^{3,31} which has an Arrhenius form. The hydrolysis degradation rate is estimated as:

$$k_H = A_H (rh_{eff})^n \exp\left(-\frac{E_H}{k_B \times T_m}\right), \quad (1)$$

where k_B represents the Boltzmann constant (8.62×10^{-5} eV/K) [Correction added on 06 March 2024, after the first online publication: In the preceding sentence ' 8.62×10^{-5} eV/K' is changed to ' 8.62×10^{-5} eV/K'], T_m is the temperature of the module (in Kelvin) and rh_{eff} (in %) represents effective module relative humidity calculated using the method proposed by Koehl et al.³² A_H is the pre-exponential constant (4.91×10^{-7}), n is a model parameter that indicates the impact of relative humidity on power loss (considered as 1.90) and E_H (0.74 eV) is the activation energy required for degradation. Values of A_H , n and E_H are used following Kaaya et al.¹⁵

2.2.2 | Thermal degradation

Thermal degradation is driven by the cyclic stress induced due to module temperature variation. We estimate the thermal degradation according to the Coffin–Manson relationship.¹⁵ The effect of the temperature cycle depends on the maximum module temperature. Thermal degradation, which has an Arrhenius form, is estimated as follows:

$$k_{Tm} = A_{Tm} (\Delta T)^\theta C_N \exp\left(-\frac{E_{Tm}}{k_B \times T_U}\right), \quad (2)$$

where k_B represents the Boltzmann constant (8.62×10^{-5} eV/K), ΔT is the difference in maximum and minimum module temperature (in Kelvin), T_U (Kelvin) represents maximum temperature in the module and C_N is the cycling rate. θ is a constant that denotes the effect of temperature difference on power degradation (considered as 2.24), A_{Tm} (considered as 2.04) is the pre-exponential constant and E_{Tm} (0.43 eV) is the activation energy for degradation.¹⁵

2.2.3 | Photodegradation

Photodegradation in PV modules is caused due to high temperature, relative humidity and exposure to UV radiation. Photodegradation, which has an Arrhenius form, can be estimated as¹⁵

$$k_p = A_p (UV)^X \left(1 + rh_{eff}^X\right) \exp\left(-\frac{E_p}{k_B \times T_m}\right). \quad (3)$$

UV represents UV radiation (kW/m^2). We obtain UV irradiance following the method by Wald.²⁶ E_p (0.45 eV) represents the activation energy required for degradation due to photoreaction, and X (considered as 0.63) is a model parameter that denotes the impact of UV radiation on degradation. A_p (considered as 71.83) represents the pre-exponential constant. rh_{eff} represents effective module relative humidity, k_B is the Boltzmann constant ($8.62 \times 10^{-5} \text{ eV/K}$) and T_m is the module's temperature (in Kelvin). We consider the values for all the constants mentioned in Kaaya et al.¹⁵

2.3 | Degradation modes

Delamination, internal circuit failure, cell ribbon corrosion and encapsulant discoloration are the dominant modes of module degradation both in terms of severity and occurrence probability.¹⁷ These modes occur under the influence of several degradation mechanisms, either independently or in synergy. Delamination and cell ribbon corrosion are caused by hydrolysis and thermal degradation mechanisms.⁶ Encapsulant discoloration in modules occurs due to hydrolysis degradation and photodegradation precursors. Internal circuit failure is formed solely due to changes in thermal cycling rate. Since the rate by which a module degrades due to a particular mode can significantly vary in different modules, we use normal distribution of the mode to model them empirically. We consider the mean value of normal distribution for delamination, encapsulant discoloration and internal circuit failure as 4.11%/year, 0.6%/year and 6.5%/year, respectively, as mentioned in the literature.^{31,33} We assume the mean value of normal distribution for cell ribbon corrosion to be 1%/year. These distributions indicate the mean annual degradation rate in the modules due to a particular mode. We model the above-mentioned degradation modes according to the following equations:

$$\text{Delamination} = \text{mean value} * \frac{(1 + k_H)(1 + k_{Tm}) - 1}{(1 + k_{H-base})(1 + k_{Tm-base}) - 1}, \quad (4)$$

$$\text{Encapsulant discoloration} = \text{mean value} * \frac{(1 + k_H)(1 + k_p) - 1}{(1 + k_{H-base})(1 + k_{p-base}) - 1}, \quad (5)$$

$$\text{Cell ribbon corrosion} = \text{mean value} * \frac{(1 + k_H)(1 + k_{Tm}) - 1}{(1 + k_{H-base})(1 + k_{Tm-base}) - 1}, \quad (6)$$

$$\text{Internal circuit failure} = \text{mean value} * \frac{(1 + k_{Tm}) - 1}{(1 + k_{Tm-base}) - 1}, \quad (7)$$

where k_H , k_{Tm} and k_p are hydrolysis degradation, thermal degradation and photodegradation. k_{H-base} , $k_{Tm-base}$ and k_{p-base} are hydrolysis degradation, thermal degradation and photodegradation for the baseline scenario. We assume average module temperature to be 47°C , minimum and maximum module temperature as 0°C and 60°C , respectively, and ambient relative humidity to be 60% in the baseline scenario. The annual UV radiation is assumed to be 80 kW/m^2 in the baseline scenario. The values for these baseline scenarios are used following previous work.^{34,35}

2.4 | Weighted average degradation rate

We model the weighted average degradation rate using the probability of occurrence of the above-mentioned degradation modes for specific climate types. We calculate the occurrence probability of each mode for three climate types: hot and humid, moderate and desert using the extensive field data collected for different regions of the world and reported by Jordan et al.¹⁷ For each type of climate, the average frequencies of occurrence of all degradation modes have been derived from Jordan's data¹⁷ (appendix tab. A1) to work as occurrence probability for modules installed both pre-2000 and post-2000. Refer to the [Supporting Information](#) for detailed method and example used to obtain the occurrence probability values in Table 2.

We use the Köppen climate classification³⁶ for Australia and separate Australia's climate zones into hot and humid, moderate and desert climate zones since the frequency of the degradation modes is presented only for these climate zones.¹⁷

TABLE 2 The occurrence probability of degradation modes depending on the climate type.

Degradation modes	Moderate	Desert	Hot and humid
Encapsulant discoloration	0.6128	0.5491	0.2741
Delamination	0.2024	0.0492	0.1321
Cell ribbon corrosion	0.0251	0.0907	0.1592
Internal circuit failure	0	0.0093	0.0072

Note: The occurrence probabilities have been calculated from the frequency of occurrences of the degradation modes from the data reported by Jordan et al.¹⁷

$$\text{Total degradation rate} = \sum \frac{\text{Degradation rate of a specific mode}}{\text{Probability of occurrence}} \quad (8)$$

We estimate the power output of mono-Si modules for each grid point using the PVWatts model³⁷ to understand the impacts of module degradation. We consider the nominal rating of the module to be 500 W and the temperature coefficient of the mono-Si module to be $-0.47\%/^{\circ}\text{C}$. To estimate the changes in power output due to module degradation, we assume the power generation capacity of the modules remains constant in the future and is not affected by climate change. The future generation capacity is considered similar to the historical generation across Australia. We only consider the changes in future degradation rates to understand the impacts of future module degradation. We estimate the future changes in LCOE to understand the economic implications of module degradation. LCOE changes reflect the changes in the electricity price of a PV plant during its lifetime due to changes in power generation capacity. The LCOE is estimated as follows:

$$\text{LCOE} = \frac{\text{Lifecycle cost (\$)}}{\text{Lifetime energy production (kWh)}} = \frac{\sum_{t=0}^n (I_t + O_t + M_t + F_t)/(1+r)^t}{\sum_{t=0}^n E_t/(1+r)^t} \dots \quad (9)$$

I_t represents the capital costs (considered as 857\$/kW), O_t and M_t are the operation and maintenance costs (considered as 14.1 \$/kW), F_t is the fuel cost (approximated as 0 for PV), E_t is the annual energy generated for the year t and r is the discount rate assumed as 7.5%. We assume the lifetime of PV modules to be 30 years. We consider the values for the different costs and discount rate according to the renewable power generation cost in the 2021 report by IRENA.³⁸ We assume the lifetime of the module and lifecycle costs remain constant during the plant lifetime to isolate the influence of module degradation on LCOE.

2.5 | Significance test

We determine the statistical significance of the changes by performing Student's t -test at a 5% significance level. We perform Student's t -test at each grid point to examine the significance of the mean change in each climate projection ensemble member individually. We represent the significance test results according to the convention proposed by Tebaldi et al.³⁹ The results are represented as regions with significant agreement, significant disagreement and insignificant agreement. When at least 50% of the ensemble members denote a significant change at a grid point, with a minimum of 70% of those members agreeing on the direction of change, it is considered a change with significant agreement and is indicated with colour and stippling. When less than 50% of the ensemble members are significant at a grid point, they are considered insignificant and are indicated with colour. When at least 50% of the ensemble members denote

significant change at a grid point, with less than 70% of them agreeing on the direction of change, they are considered as regions with significant disagreement and are represented in white.

3 | RESULTS

3.1 | Impacts of future climate change on module degradation

Previous studies have projected an increase in module temperature over Australia and subsequent changes in other variables (such as irradiance, wind speed and relative humidity) that trigger module degradation.^{40,41} Total and partial failure of the modules occurs due to the concurrent occurrence of different degradation modes. The probability of occurrence of these modes is dependent on the regional climate. In this study, we weigh the occurrence probabilities of the degradation modes to model the total degradation rate referred to as the 'weighted average module degradation rate' (refer to Section 2).

To assess the impacts of climate change on module degradation, we estimate the future changes in the weighted average module degradation rate between the historical period (1976–2005) and the future period (2030–2059) (shown in Figure 2). We observe that mono-Si modules tend to have higher weighted average degradation rates for regions with a hot and humid type of climate (Northern Australia; up to 0.3%/year). The weighted average degradation rates are less pronounced in Central Australia (Figure 2a). This region has an arid type of climate with lower humidity. It should be noted that the historical degradation rates reported by Jordan and Kurtz⁵ and Copper et al.⁴² are higher than the values obtained in our results. The main reason behind this is our model includes only the four main degradation modes occurring in mono-Si modules. Modules can also degrade due to other modes like PID, glass breakage, cell cracks and junction-box failure that have not been considered here.

Our results predict an increase in the total degradation rate in the future under both scenarios (Figure 2b,c). Changes in the weighted average degradation rate are more pronounced for Northern Australia under RCP2.6 (up to 0.06%), with the smallest increases in the central region. However, under RCP8.5, we observe higher changes in the weighted average degradation rate in the northern and eastern regions of the country (up to 0.1%/year). We observe that the weighted average degradation rates will increase further by the end of the century. The spatial pattern of future degradation rate is consistent; however, the magnitude of the changes almost doubles (Figure S2).

We estimate the impacts of accelerated future weighted average degradation rates on power output and assess the economic impacts of these changes. To analyse the sensitivity of module degradation on power generation capacity, we do not consider changes in future power output due to climate change. The future power generation capacity across Australia is considered equivalent to the historical

Weighted Average Degradation Rate of mono-Si Modules

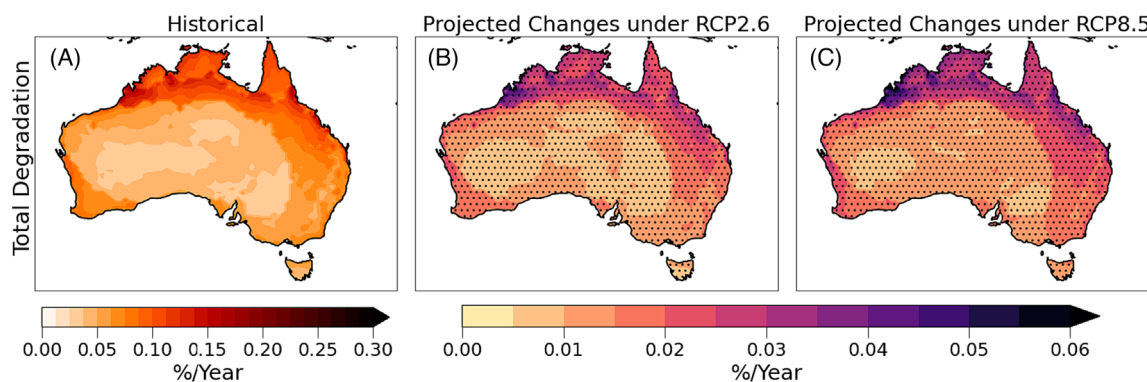
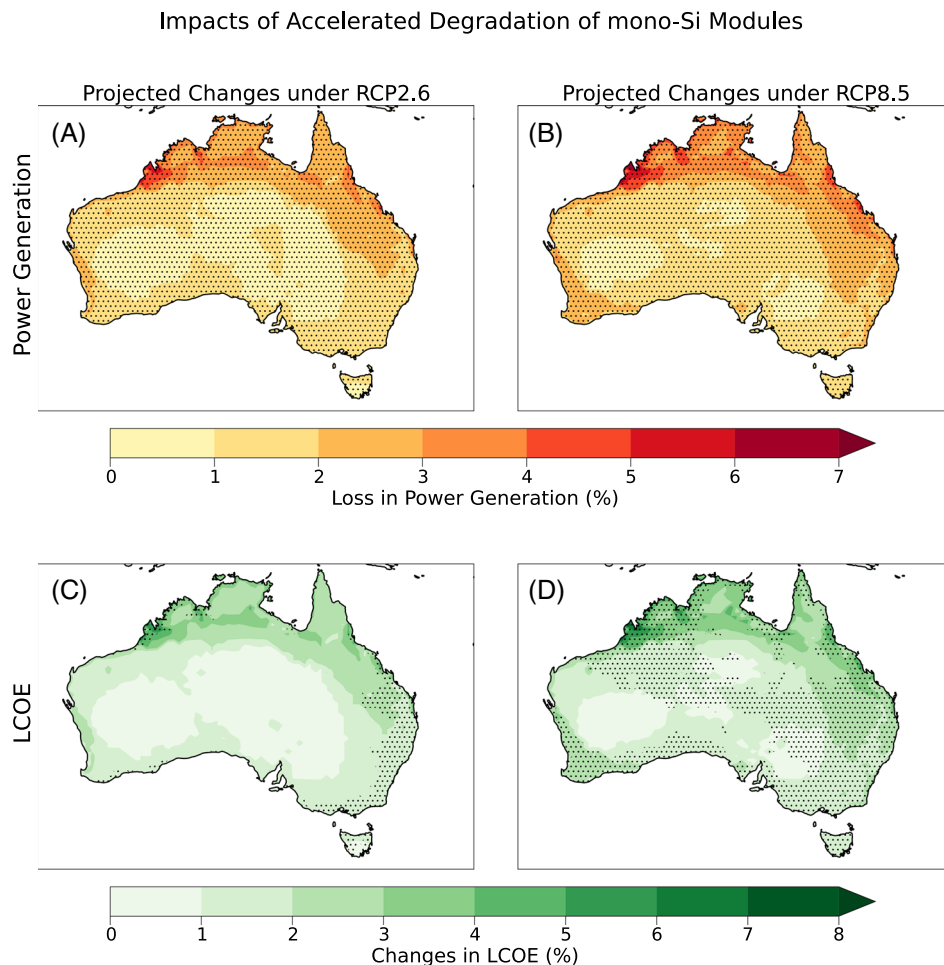


FIGURE 2 The weighted average degradation rate of mono-Si modules. Panel (a) shows the total degradation rate of mono-Si modules during the historical time period (1976–2005). Panel (b) and (c) show future changes (2030–2059) in total degradation rates of mono-Si modules under RCP2.6 and RCP8.5 scenarios with respect to the historical period. Significant changes in the future are represented by stippling (see Section 2.5).

FIGURE 3 Impacts of accelerated mono-Si module degradation. Panel (a) and (b) show future changes (2030–2059) in power generation due to accelerated module degradation under RCP2.6 and RCP8.5 future scenarios, respectively, with respect to the historical period (1976–2005). Panel (c) and (d) show future changes (2030–2059) in LCOE due to accelerated module degradation under RCP2.6 and RCP8.5, respectively, with respect to the historical period (1976–2005). Significant changes in the future are represented by stippling (see Section 2.5).



value. The future changes in the power output of mono-Si modules due to future degradation rates under RCP2.6 and RCP8.5 are shown in Figure 3a,b.

In the future, we expect power loss throughout the country due to enhanced module degradation. Our results project ~6% decrease in power output in Northern Australia under RCP2.6 (Figure 3a) and

up to 8.5% under RCP8.5 (Figure 3b). Power loss is minimum in Central Australia and Tasmania (up to 2%) under both future scenarios. It is worth noting that hot and humid and moderate climate types are more vulnerable to power loss in the future. As expected, the highest power loss is in the regions with high degradation rates (Figure 2).

Changes in the future power generation capacity of the modules due to degradation can have large economic impacts. Due to an increase in future power loss, the cost of energy produced is projected to increase in the future (relative to its current trend). We use the LCOE metric to estimate additional costs incurred in the future. Note that we do not consider future changes in solar resources or increase in future PV capacity when estimating LCOE changes to isolate the impact of module degradation on future costs. Our results show an increase in LCOE throughout the country under both future scenarios. We observe maximum changes in LCOE in the north-western part of the country up to 6.5% under RCP2.6 (Figure 3c) and up to 10% under RCP8.5 (Figure 3d). This means that energy will cost an additional 8 cents/kWh in those regions. The cost of energy in Eastern Australia is projected to increase up to 3 cents/kWh (up to 5%) in the future from the present rate (~70 cents/kWh). As

expected, Central Australia is projected to undergo the least changes in future LCOE (up to 2%). We project that by the end of the century, the energy cost shall increase up to 8% under the RCP2.6 scenario (Figure S3). Under RCP8.5, we predict a 2% increase in LCOE near the central regions of Australia, up to 8% in the east and south and up to 15% in the north due to an increase in future power loss (Figure S3).

3.2 | Degradation of mono-Si modules: drivers and projected future changes

3.2.1 | Degradation mechanisms

Temperature, relative humidity and UV radiation trigger hydrolysis degradation, thermal degradation and photodegradation mechanisms

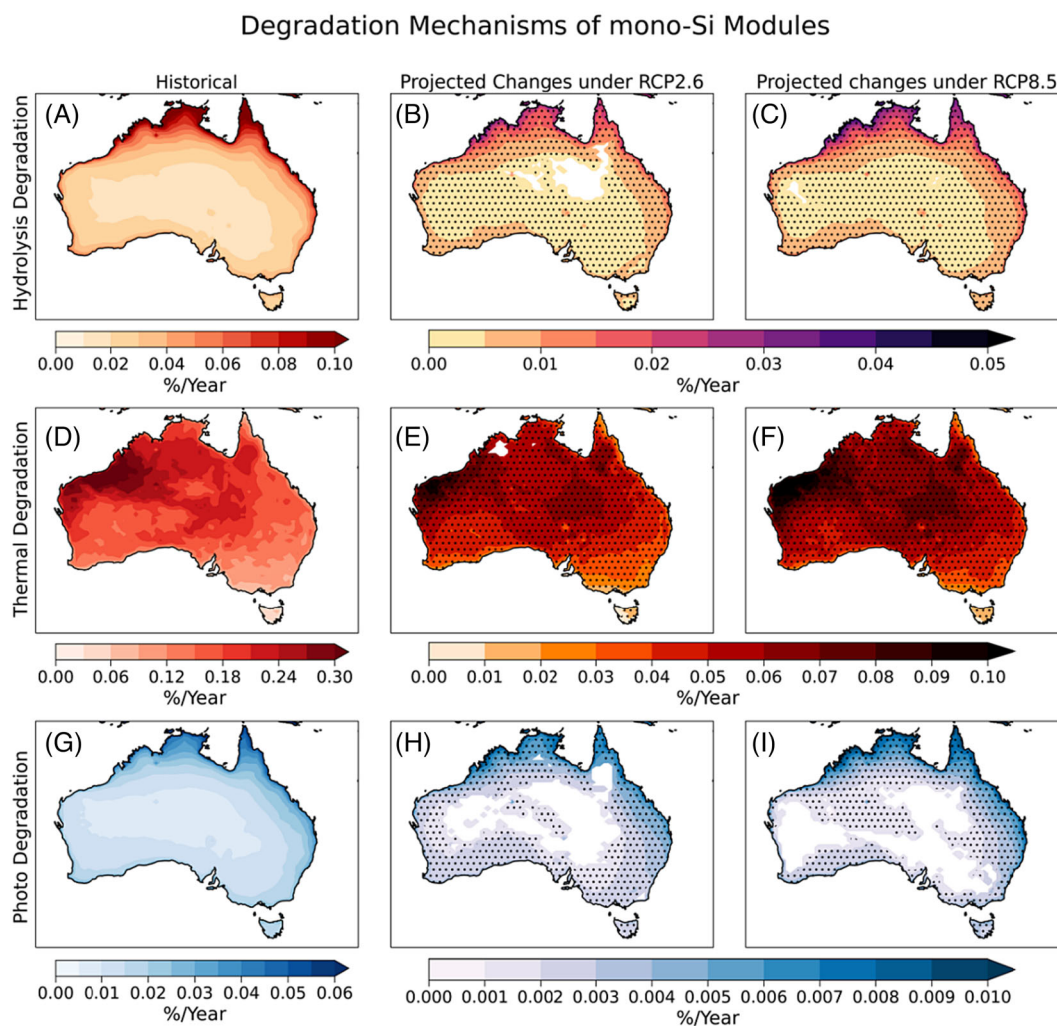


FIGURE 4 Precursors of mono-Si module degradation. Panel (a) represents hydrolysis degradation for the historical time period (1976–2005). Panels (b) and (c) represent the future changes (2030–2059) in hydrolysis under RCP2.6 and RCP8.5, respectively, concerning the historical time period. Panel (d) represents thermal degradation for the historical time period, and future changes in thermal degradation under RCP 2.6 and RCP 8.5 concerning the historical time period are shown in panels (e) and (f), respectively. Photodegradation during the historical period is shown in panel (g). Future changes in photodegradation under RCP2.6 and RCP8.5 concerning historical time period are shown in panels (h) and (i), respectively. Significant changes in the future are represented by stippling (see Section 2.5).

in PV modules. These degradation mechanisms act as precursors and initiate several degradation modes. Understanding these mechanisms and modes individually is essential to understand how regional climate influences them and which mode dominates in different parts of Australia. We start by analysing the degradation mechanisms for mono-Si modules (see Section 2; Figure 1) for the historical period (1976–2005) and estimate the future changes (2030–2059) under RCP2.6 and RCP8.5 scenario (Figure 4).

We observe that the thermal degradation mechanism dominates Australia historically, followed by hydrolysis degradation and photodegradation. Thermal degradation is dependent on the thermal cycling of the modules and the daily maximum module temperature. During the historical period, we observe the highest thermal degradation in West and Central Australia (up to 0.3%/year) (Figure 4d). These regions have a desert climate with high temperatures all year around. With lower daily maximum temperatures, southern regions of Australia have lower thermal degradation rates. The next most dominant mechanism is hydrolysis. Hydrolysis degradation is driven by relative humidity and air temperature. Hence, tropical regions of Australia have a high impact due to hydrolysis. Results show that degradation due to the hydrolysis mechanism is highest in Northern Australia (up to 0.10%/year) (Figure 4a). Hydrolysis leads to delamination of the polymers and corrosion of modules due to moisture ingress in the back sheet. PV modules installed in tropical regions experience high temperatures, humidity and UV irradiance and tend to degrade faster due to photodegradation than installations in other climate types. Photodegradation is highest in Northern Australia (up to 0.06%/year) during the historical period (Figure 4g).

The thermal degradation rates are projected to escalate under RCP2.6 and RCP8.5 scenarios. As shown in Figure 4e,f, the highest changes in thermal degradation rates are expected in the West (>0.10%/year) with a moderate increase in Northern and Central Australia. We expect such increments in thermal degradation rates due to increased module temperature in the future.⁴⁰ Under a future warmer climate, we expect the hydrolysis degradation mechanism to increase under both RCP2.6 and RCP8.5 scenarios. These changes are more pronounced for coastal regions of Australia (up to 0.05%/year), with negligible changes in the central part of Australia (Figure 4b,c). Our results show that photodegradation increases under both the future scenarios. We expect the biggest changes in future photodegradation rates in the North under both scenarios (up to 0.01%/year; Figure 4h,i). Results indicate higher future changes under RCP8.5 than RCP2.6 for all the degradation mechanisms.

The spatial mean and distribution pattern of historical degradation mechanisms agree with the previous research.¹⁶ Ascencio-Vázquez et al.¹⁶ show that historically Northern Australia has the highest degradation rates for hydrolysis degradation (0.125%/year) and photodegradation mechanism (~0.1%/year), and thermal degradation rate (~0.4%/year to 0.5%/year) is highest in Western and Central Australia. Our results show a similar pattern for the historical period. However, the spatial means for hydrolysis degradation, thermal degradation and photodegradation mechanisms were obtained only from 2016 to 2018 by Ascencio-Vázquez et al.¹⁶ Therefore, due

to the difference in the time periods considered in the study, we observe small differences in the historical mean degradation rates for the mechanisms in our results.

3.2.2 | Degradation modes

We empirically model the degradation modes for mono-Si modules induced due to one or more degradation precursors (refer to Section 2). Figure 5 shows rates of delamination, cell ribbon corrosion, internal circuit failure and encapsulant discoloration for the historical period and projected changes in the future under RCP2.6 and RCP8.5. Delamination is one of the dominant modes of degradation caused due to moisture ingress and high temperature leading to loss of adhesion between the polymer and the cells or cells and the front glass.¹⁷ Further, moisture retention in the module leads to corrosion. Moisture retention in the module is a significant safety concern and causes power loss due to increased leakage current.^{43,44} Internal circuit failures (or solder bond failures) occur due to thermal stress and changes in the maximum and minimum temperature of the modules during each cycle.⁴⁵ Overexposure of the modules to high UV doses can cause the module to change its colour to yellow or brown, known as encapsulant discoloration. Encapsulant discoloration is a dominant degradation mode in modules installed for over 10 years.¹⁷

On comparing the degradation rates of the several modes historically, we observe that internal circuit failure has the highest values followed by delamination, cell ribbon corrosion and encapsulant discoloration during the historical period (Figure 5a,d,g,j). Internal circuit failure is highest in the West (up to 1.2%/year) during the historical period (Figure 5j). They are mainly caused due to thermal cycling of the modules and hence tend to have high values for desert and arid regions of Australia. Our results show that historically delamination of modules is highest in the West (>0.5%/year) and decreases towards the South of Australia (up to 0.25%/year). Australia's western, northern and central regions experience higher cell ribbon corrosion historically than the South (Figure 5g). Thermal and hydrolysis degradation mechanisms influence delamination and cell ribbon corrosion occurrences in modules and are higher in regions with high temperature and humidity. Encapsulant discoloration is higher in the northern coastal regions of Australia (up to 0.2%/year) during the historical period (Figure 5d). It is caused by hydrolysis degradation, thermal degradation and photodegradation mechanisms. Hence, northern regions with high humidity, temperature and UV dose have higher encapsulant discoloration rates than other parts of Australia. Even though the arid and semi-arid regions have high temperatures and UV irradiance, they have lower encapsulant discoloration due to lower humidity.

Note that even though a region has a higher value for a specific degradation mode, it might have less influence on the overall degradation rate due to its lower occurrence probability. For example, Central Australia experiences moderately high internal circuit failure degradation rates (up to 1%/year), delamination degradation rates (up to 0.35%/year) and cell ribbon corrosion degradation rates (up to 0.09%/year) historically but has the lowest weighted average degradation

Degradation Modes of mono-Si Modules

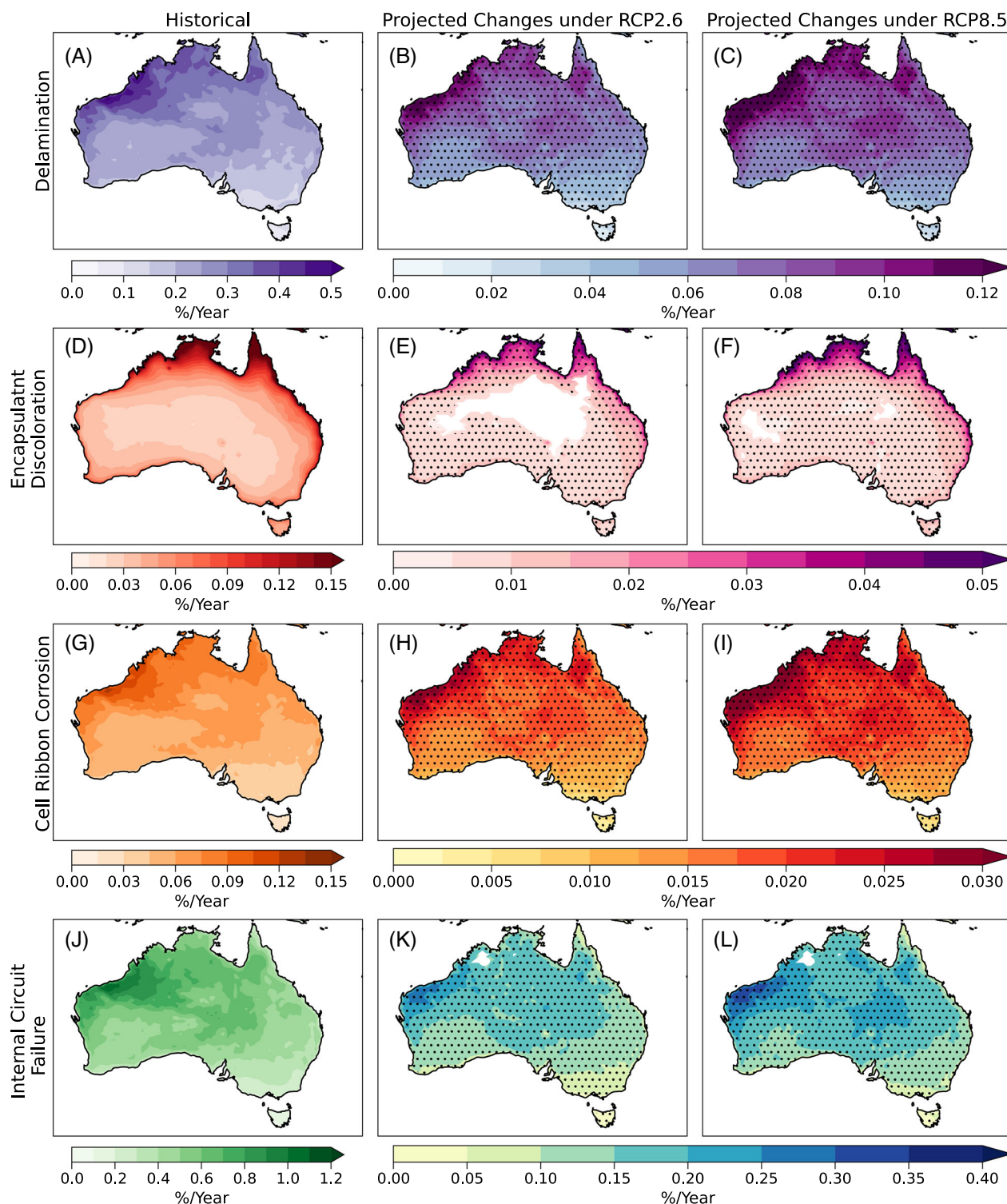


FIGURE 5 Degradation modes in mono-Si modules. Panel (a) represents delamination during the historical time period (1976–2005). Panels (b) and (c) represent future changes (2030–2059) in delamination under RCP2.6 and RCP8.5, respectively, with respect to the historical time period. Encapsulant discoloration for the historical time period is depicted in panel (d), and future changes in encapsulant discoloration under RCP2.6 and RCP8.5 with respect to the historical period are shown in panels (e) and (f), respectively. The degradation of mono-Si modules due to cell ribbon corrosion in the historical period is shown in panel (g), and future changes in the cell ribbon corrosion under RCP2.6 and RCP8.5 concerning the historical period are shown in panels (h) and (i). Panel (j) shows degradation due to internal circuit failure during the historical period. Future changes in internal circuit failure under RCP2.6 and RCP8.5 concerning the historical period are shown in (k) and (l), respectively. Significant changes in the future are represented by stippling (see Section 2.5).

rate due to lower occurrence probability of these modes (Figure S1). We consider the occurrence probabilities from Jordan et al.¹⁷ On analysing the degradation rates of encapsulant discoloration, delamination, internal circuit failure and cell ribbon corrosion weighted by their occurrence probability, we observe that delamination is the dominant mode of degradation in mono-Si modules for all climate types in Australia (Figure S1).

Our results project an increase in the degradation rate of internal circuit failure in the future due to climate change. We expect a higher increase in future internal circuit failure in Western Australia under RCP2.6 and RCP8.5 scenarios, with moderate increases in the central regions of Australia and smaller increases in the South (Figure 5k,l). Similarly, delamination is expected to increase in the future in Australia (Figure 5b,c). Changes in delamination are expected to be higher in the North and West (>0.12%/year), moderate in the central regions (up to 0.08%/year) and least in the South (up to 0.04%/year). Additionally, we observe that these regions with high cell ribbon corrosion are expected to have more pronounced future changes under both scenarios (Figure 5h,i). Our results also predict an increase in encapsulant discoloration in the northern coastal regions of Australia (up to 0.05%/year) under RCP2.6 and RCP8.5 scenarios. Future changes in encapsulant discoloration are less prominent for other regions in Australia (Figure 5e,f). Our results suggest that future changes in the degradation modes are more severe under the RCP8.5 future scenario when compared to RCP2.6 (up to 1.5 times). Even though the spatial pattern of the future changes in degradation modes are similar for RCP 2.6 and RCP8.5, we observe that the magnitude of the rates intensifies under RCP8.5.

4 | DISCUSSION

Mitigation of climate change is closely related to the uptake of renewable energy systems, and solar PV is one of the preferred sustainable energy systems. However, exposure of the PV modules to atmospheric conditions triggers degradation leading to power loss and reduced lifetime. Future climate change may enhance module failure and increase the need for module replacement. Therefore, it is crucial to understand the driving mechanisms and role of climate change on future module degradation. This paper assesses the impacts of climate change on mono-Si module degradation in Australia under different future scenarios.

The various degradation mechanisms and modes for mono-Si modules are studied using the latest RCM projections over Australia. We assessed the outcomes for both the historical period and estimated the future changes based on the optimistic scenario, 'RCP2.6' and the pessimistic scenario 'RCP8.5'. Our results project an increase in the weighted average degradation rates under both future scenarios. The regions showing statistically significant future changes in module degradation are denoted in stippling. These high future degradation rates can lead to significant increase in power loss (up to 12%) and escalate the LCOE (~10% to 12%) by 2059. Therefore, it is expected that the operational and proposed PV plants with mono-Si

modules will incur substantial losses in the future, not only due to module degradation but also due to frequent module replacement due to module failure. Projected increase in occurrence of extreme weather events⁴¹ like cyclones, hailstorms, high wind gusts and floods can cause material damage and accelerate module degradation leading to frequent module replacements in the future. Furthermore, we have shown the effect of future degradation rate on LCOE assuming the future meteorological parameters and costs remain similar to the historical period. However, we expect a reduction in the future PV yield over Australia due to climate change,⁴⁰ and previous studies have forecasted a future reduction in the different cost parameters for PV.⁴⁶ It will be interesting to include these variations in LCOE to obtain more robust results from future PV economy point of view.

We identify that thermal degradation is Australia's main degradation precursor, followed by hydrolysis degradation and photodegradation. This means that future research should focus on reducing the extent of thermal cycling in modules by incorporating sophisticated cooling techniques⁴⁷ and improving their material properties to minimize thermal degradation. We predict the highest increase in internal circuit failure degradation mode followed by delamination by 2059 under both future scenarios. Keeping this in mind, it is essential to focus on improving the module design to limit temperature rise of the modules. This would ensure higher power output and better lifetime of the modules. At the same time, further research into advanced module manufacturing processes is necessary to limit moisture ingress in the back sheet. It is important to note that this study analyses the mean degradation modes of widely used mono-Si modules. However, with current advancements (half-cell and shingle modules, bifacial modules including glass-glass and glass polymer modules, busbarless modules and n-type module technologies), the market has more temperature resilient modules. It is highly advisable to perform similar degradation modelling analysis along with techno-economic assessments for a wide range of PV technology in Australia. This will benefit future large-scale PV plants to choose the most cost-effective and climate resilient modules.

Our results show that the impact of climate change on PV module degradation is more pronounced under high-emission scenarios. RCP2.6 simulates a low-carbon pathway that incorporates climate protection strategies along with renewable energy integration and hence projects lower changes in module degradation. On the other hand, RCP8.5 represents business as usual with increased carbon emissions in the future that accelerate future climate change. Thus, RCP8.5 projects higher module degradation and has a detrimental effect on LCOE. It is essential to consider module degradation under different future climate scenarios to understand the climate uncertainties and its impacts due to different mitigation strategies. Also, the range of diverse projections helps in stimulating technological development pathways to ensure minimum power loss even in the worst-case scenario. This work discusses the results for the CORDEX-Australasia ensemble mean. However, each ensemble member has a unique response to future climate due to different parameterization schemes and driving GCMs. The ensemble members show a wide

range of future possibilities. Most of the members resemble similar types of future change, with few having opposite changes (see the [Supporting Information](#)). This highlights the importance of including multiple ensemble members to quantify the results confidently. Our results highlight the importance of regional climate while estimating the total degradation rate and its drivers. Our findings show that similar module technologies may have different future degradation rates when installed in different climate zones across Australia. Furthermore, our results highlight the importance of considering the probability of occurrence of a failure mode when estimating the net degradation. This work establishes a foundation for modelling future module degradation on a global scale, employing a novel approach known as weighted average degradation. This method integrates the degradation rates of individual modes and considers their occurrence probabilities, which are influenced by the climate type to which they are exposed. This research is highly beneficial for improving module lifetime through appropriate material selection and technological advancements in the manufacturing process.

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DATA AVAILABILITY STATEMENT

No new data were generated in the study. The CORDEX-Australasia data used for the analysis are available from the Earth System Grid Federation (<https://esgf-data.dkrz.de/search/cordex-dkrz/>).

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SUPPORTING INFORMATION

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